DATA ANALYSIS

Zuber Rides in Chicago

Exploratory data analysis and statistical hypothesis testing on ride behaviour and the impact of weather on trip duration

Python · Pandas · Matplotlib · SciPy · SQL

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CONTEXT

A new ride-sharing company in Chicago

Zuber is a new ride-sharing company launching in Chicago. Before entering the market, it needs to understand how the city already moves.

Project goal

Map ride patterns across the city, identify trends among taxi companies and the busiest neighbourhoods, and investigate the impact of weather on trip duration.

1

Exploratory analysis

Dataset quality, companies with the most rides, busiest neighbourhoods and charts.

2

Hypothesis test

Weather vs. ride duration, comparing rainy and dry Saturdays with Student's t-test.

METHODOLOGY

Data and tools

`project_sql_result_01.csv`

64 taxi companies

Total rides per company

`project_sql_result_04.csv`

94 drop-off neighbourhoods

Average rides ending in each neighbourhood

`project_sql_result_07.csv`

1,068 rides

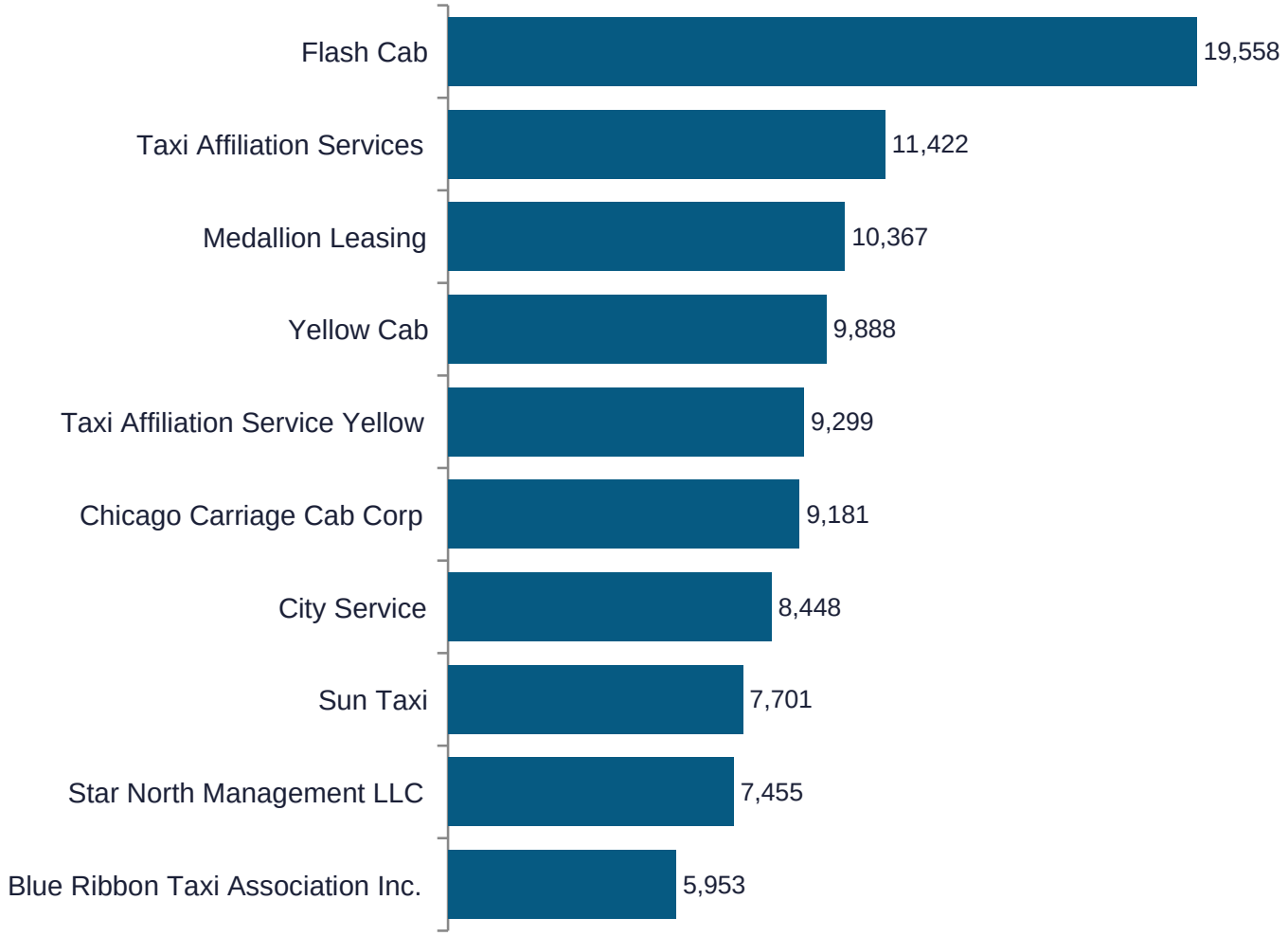
Loop → O'Hare, with weather and duration

Data quality

- No missing values in any of the three datasets
- 197 duplicate records in the rides dataset, kept because there is no unique trip ID
- start_ts column converted to datetime

EXPLORATORY ANALYSIS

Top 10 taxi companies



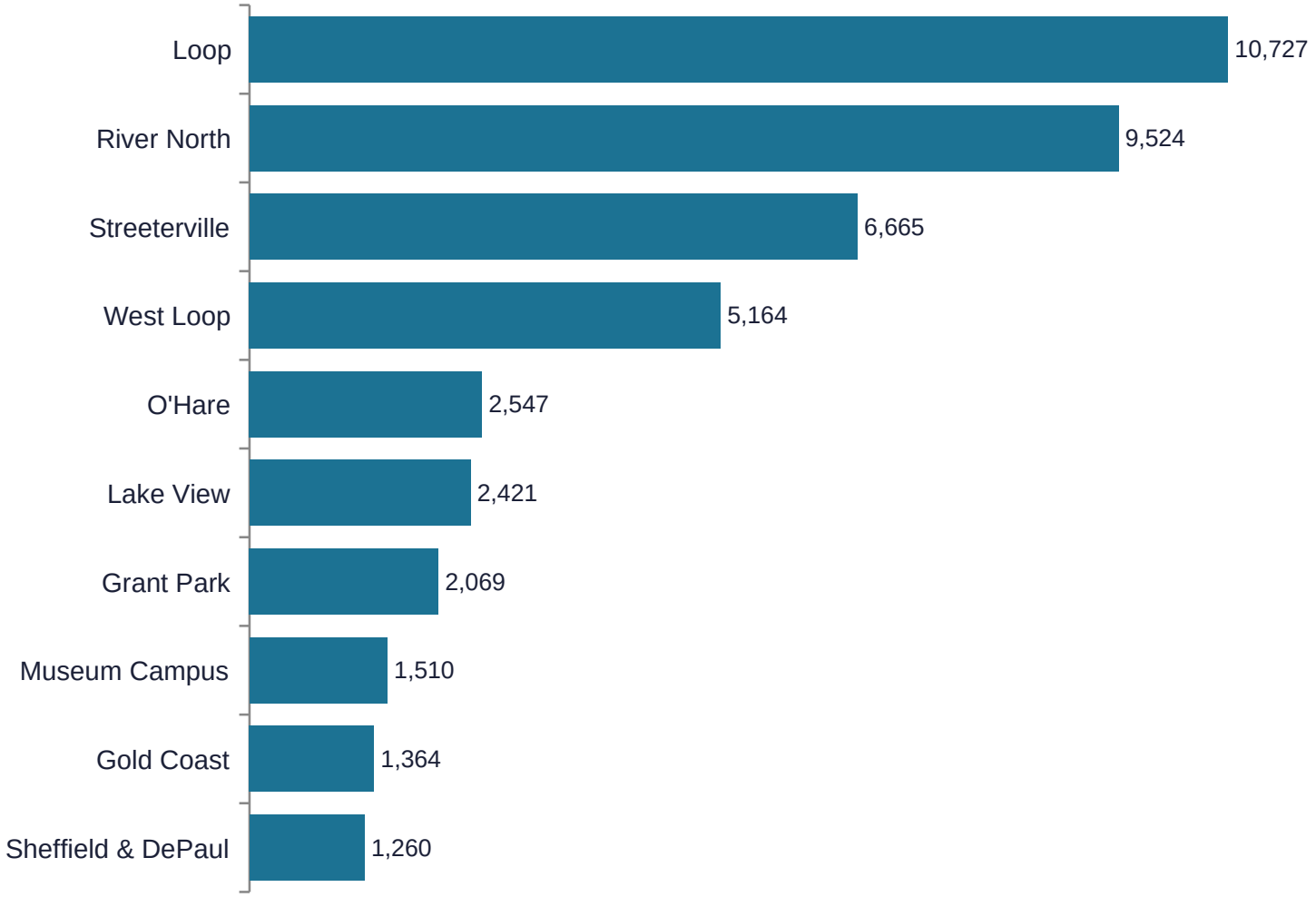
19,558

rides by Flash Cab, the clear market leader

The top three companies account for about 30% of all rides across the 64 companies analysed, a market with strong concentration at the top.

EXPLORATORY ANALYSIS

Top 10 neighbourhoods by rides



10,727

average rides ending in the Loop, the city's financial district

Loop, River North, Streeterville and West Loop hold the highest demand, reflecting central areas with strong business and tourist activity.

HYPOTHESIS TEST

Does weather affect ride duration?

Scope: rides between the Loop and O'Hare International Airport on Saturdays, comparing bad weather (rain) and good weather days.

H_0 · Null hypothesis

Average ride duration on rainy Saturdays equals the average ride duration on dry Saturdays.

H_1 · Alternative hypothesis

Average ride duration on rainy Saturdays differs from the average ride duration on dry Saturdays.

Student's t-test

independent samples, unequal variances

$\alpha = 0.05$

significance level

148 vs. 723

rides in bad vs. good weather

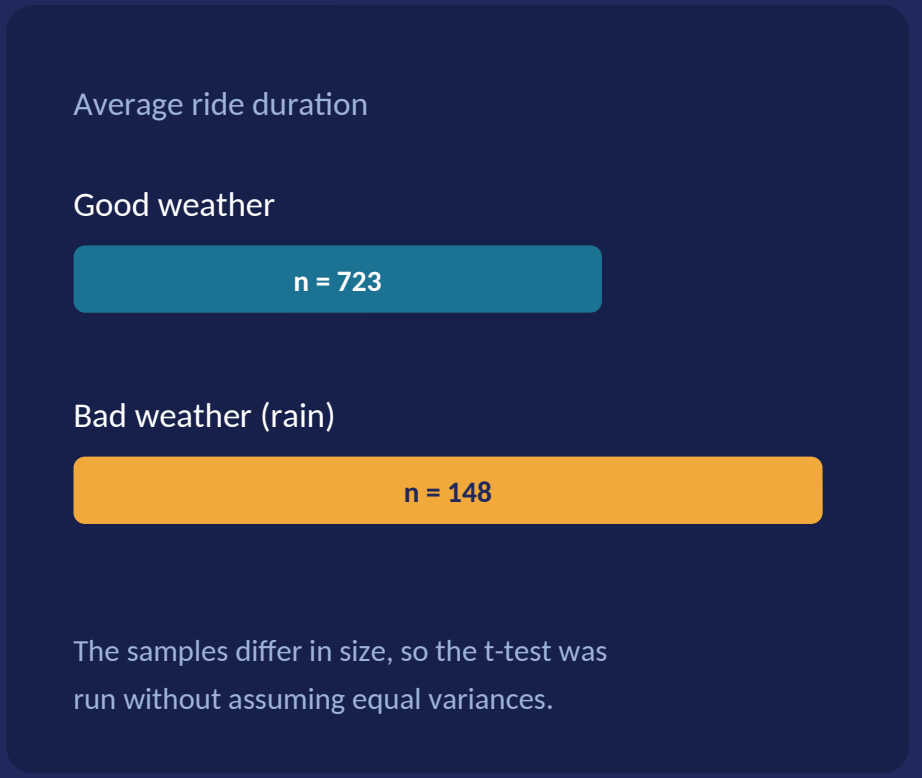
TEST RESULT

H_0 rejected: weather changes ride duration

$$p = 9.13 \times 10^{-8}$$

far below $\alpha = 0.05$

The statistical evidence shows that average ride duration between the Loop and O'Hare Airport on Saturdays is different on bad-weather days compared with good-weather days.



CONCLUSION

What this means for Zuber

1

Concentrated market

A few companies, such as Flash Cab, dominate ride volume. There is room to stand out against smaller operators.

2

Demand by area

Loop, River North, Streeterville and West Loop hold the highest demand and should guide fleet allocation.

3

Weather as a variable

Weather affects ride duration on airport routes, which suggests adjusting pricing and ETAs on rainy days.

Suggested next step: cross peak hours by neighbourhood with weather history to anticipate demand and travel-time peaks.